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Should I Stay or Should I Go Pro? Early NFL Draft Entry by NCAA FBS Underclassmen

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Abstract
Some college players face the tradeoff between continuing their education and entering the draft and forfeiting college eligibility. Little empirical research focuses on this "stay or go pro" decision. We analyze early draft entry decisions made by college football underclassmen with remaining eligibility over the 2007/08-2018/19 seasons. Regression results support both the human capital explanation for early draft entry in that players acquiring more human capital in the previous season by playing in more games increases the likelihood of early entry and the option value perspective in that playing on a more successful team increases the likelihood of early entry.

Keywords
entry draft, football, intertemporal choice, returns to education

Introduction

All professional sports leagues in North America conduct annual amateur entry drafts, generating monopsony power for leagues. While all domestic amateur athletes

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must pass through entry drafts, draft eligibility rules provide flexibility in terms of the timing of entry. Players can choose to exhaust their 4 years of eligibility before entering the draft, or alternatively choose to enter the draft early, before their college eligibility ends, with some restrictions. The option of declaring early for the draft, uncertainty about where a player will be selected, and the economic consequences of selection lower in the draft generate an interesting setting for analyzing high-stakes, intertemporal labor supply decisions.

Previous research primarily focused on entry draft and postdraft outcomes in terms of round of selection, starting salary, and career length, motivated by economic models of firm hiring, using data on players in the National Basketball Association (NBA) (Ichniowski & Preston, 2017; Motomura et al., 2016) and the National Football League (NFL) (Berri & Simmons, 2011; Bollinger & Hotchkiss, 2003; Grier & Tollison, 1994; Hendricks et al., 2003; Kitchens, 2015; Pitts & Evans, 2018; Rosen & Olbrecht, 2020). While these papers analyze how early draft entrants fared in sports labor markets after declaring early compared to players who competed in college for 4 years, none empirically analyzed the decision to enter the draft early from the perspective of underclassmen facing a "stay or go" decision.

Two previous papers analyzed early draft entry decisions from a theoretical perspective. Groothuis et al. (2007) developed a model explaining the decision by college players to declare early for the NBA draft. This model emphasized the importance of human capital investment in decisions to declare early or remain in college, as well as the role of forgone professional earnings. Arel & Tomas (2012) developed a model based on the idea that the decision to declare early resembles the decision to exercise an American put option by the option owner in financial markets. Arel & Tomas (2019) extended the put option model to include a salary "racheting" effect. Both models identify factors associated with players' decisions to enter early and inform the empirical work in this paper. No previous research tests the predictions made by these competing models.

We address this lack of evidence through an empirical analysis of the decision by college football players to declare early for the NFL entry draft. We analyze the early draft entry decisions of NCAA Football Bowl Subdivision (FBS) underclassmen over the 2007/08-2018/19 college football seasons. Eight hundred and thirty-six eligible underclassmen declared early for the NFL draft over this period. We use an exact matching approach to construct a comparison group for these early entrants from FBS underclassmen who were eligible to declare early but did not. We match players based on their high school Rivals.com star ranking, position, conference, and height.

Results from a linear probability model (LPM) applied to a dependent variable equal to one if a player declared early for the draft show that the probability a player declared early increases by about 3% to 5% with each additional game played in the season before the draft, supporting the predictions of the human capital early entry decision model developed by Groothuis et al. (2007). The results also show that each additional win by the player's team in the season

before the draft increases the probability of early entry by about 2.5%, supporting the predictions of the option value model developed by Arel & Tomas (2012). Results from a robustness check focusing only on quarterbacks based on a model that contains a rich set of player level performance variables confirm these results.

Our paper contributes to the literature by undertaking the first empirical analysis of early draft entry decisions from the perspective of underclassmen athletes. All previous research on early draft entry focused on the professional careers of early entrants compared to players who exhausted their college eligibility, but did not compare early entrants to other underclassmen eligible for early entry who decided to stay in college. This approach also allows us to conduct the first empirical test of the predictions of the two models explaining early draft entry decisions.

Literature Review

A number of previous studies analyzed the implications of the draft from the demand side in the professional football, basketball, and baseball player markets. Many focused on the NFL draft using a variety of approaches to analyze the ability of teams to appropriately value talent and draft players, a labor demand topic focused on firm hiring decisions (Berri & Simmons, 2011; Bollinger & Hotchkiss, 2003; Grier & Tollison, 1994; Hendricks et al., 2003; Massey & Thaler, 2013). These papers focused on draft outcomes and subsequent professional performance, because NFL draft data and player performance data can be easily obtained.

While this approach can uncover interesting economic aspects of early draft entry, as well as insight into team hiring decisions, it cannot provide insight into decisions made by college players to enter drafts early or remain in college an additional year, since outcomes for players who stayed in college an additional year are not analyzed. We address this omission in our empirical analysis.

Two papers addressed decisions by college players to enter the draft or remain in college from a theoretical perspective.[1] No previous research empirically analyzed the decision to stay in college or turn pro. While this decision can be influenced by many factors, these two theoretical models provide clear predictions that can be empirically tested.

Groothuis et al. (2007) developed a model of a college player's decision to declare early for the draft from a human capital investment perspective. Economists have long recognized the importance of human capital in economic models (Folloni & Vittadini, 2010). Empirical research on human capital assumes that education and work experience represent key proxy variables for human capital (Slottje, 2010). The model developed by Groothuis et al. (2007) compared expected total career earnings for a player who stays in college one more year to expected total career earnings for a player who declares early under the assumption that the player will earn a salary less than the marginal revenue product produced during the rookie contract because drafts generate monopsony power for professional teams. The model accounts for

player ability, increase in human capital should the player remain in college for one additional year, and how much less than the marginal revenue product produced a player will earn during the rookie contract period.

The model predicts that the larger the increase in ability from staying in college an additional year, the fewer underclassmen will declare. It also predicts that the smaller the monopsony exploitation of rookies by professional teams, the fewer early entrants. The model developed by Groothuis et al. (2007) also makes predictions about the impact of factors like the length of the rookie contract, the expected increase in human capital over another year playing in college, and expected career length, on the decision to declare early. The model focuses on players' human capital acquisition from an additional year of college play relative to forgone professional earnings lost by staying in college another year. Players who expect to gain more human capital from an additional year of college play will be less likely to declare early for the draft.

In order to link the predictions of the model developed by Groothuis et al. (2007) to our empirical analysis, we assume that college athletes increase their human capital through a "learning by doing" process where playing in more college games increases the player's stock of human capital and playing in games against higher quality opponents also increases the player's stock of human capital. We also assume diminishing returns to human capital acquisition through this process: the expected amount of additional human capital acquired by a college player who expects to appear in every game in the next season will be less than the human capital acquired by a college player who appeared in every game this season, other things equal. This assumption also applies to additional human capital acquired by playing against high-quality opponents.

In addition to developing a model of decisions made by college athletes, Groothuis et al. (2007) also estimated wage equations using data on the player's NBA earnings and performance from 1997 and 2002 explaining observed variation in salary for players with 3 years or less experience (those playing under the rookie salary scale). Results indicate that players who entered the NBA draft early improved more and played fewer minutes in their first year in the NBA than first year players with 4 years of college experience. The empirical analysis focuses on draft outcomes and NBA salaries for players under the rookie salary scale, and not on the decision by players to declare early. The paper also develops a model of teams' hiring decisions similar to the models by Hendricks et al. (2003) and Bollinger & Hotchkiss (2003).

Arel & Tomas (2012) modeled the decision to declare early for the draft as analogous to exercising an American Put Option from the finance literature. A put option is the right to sell an asset at a predetermined price, called the strike price or exercise price. An American put option can be sold ("exercised") on or before some specific date; a European put option can only be sold on a specific date, called the maturity date. Under this analogy, players can exercise the option to sell their services to a professional team early by entering the draft, or delay exercising the option by remaining in college another year.

Owners of an American put option will exercise early for option values sufficiently "deep in-the-money" meaning the current option value falls below the value at maturity by a large enough amount. In the context of the decision to declare early for the draft, the option matures when the player exhausts all four years of college eligibility, and the option value at maturity is the player's expected career professional earnings when he plays four years in college and then enters the draft. The option value when a player declares early for the draft is the value of the players remaining eligibility.

Arel & Tomas (2012) approach emphasizes the importance of the value of an additional year in college, in terms of additional experience, training, and other forms of human capital relative to the payoff to the player in terms of his draft position this year compared to next year if he stays in college. The put option approach also emphasizes the role played by uncertainty in the player's decision. Early exercise of an American put option occurs when the underlying asset has lost enough value to make the strike price attractive. The earlier in the draft, a player is selected, the higher the strike price. In the draft context, the underlying asset is the player's remaining eligibility valued in terms of an expected increase in draft position from the current draft to the next draft. The strike price reflects the player's expected draft position in the current draft. The player cannot know exactly where he will be drafted in the current or future draft until the draft occurs, and he must decide whether to exercise the option or not before the draft occurs.

In order to link the predictions of the model developed by Arel & Tomas (2012) to our empirical analysis, we exploit a well-known empirical regularity: the higher a player is selected in each year's entry draft, the larger his salary during the period before he gains free agency in the professional labor market. Earlier draft picks earn higher salaries than later draft picks. Any observable college outcome that leads a college player to be selected higher in the upcoming draft will increase the likelihood the player declares early for the upcoming draft, holding factors related to future human capital acquisition constant.

We posit that team wins in the college season before the NFL draft represents a proxy variable for factors that affect the option value of a player's remaining eligibility. While a general perception of persistent college football team success exists, in other words, the idea that winning teams continue to win season after season, the number of team wins this season is not a strong predictor of the number of team wins next season. A two-way fixed effects panel data regression with current season wins as the dependent variable and current season strength of schedule and wins last season as explanatory variables produces a statistically significant coefficient on the lagged wins variable of just 0.24. Each additional win this season is associated with just one-quarter of a win in the following season, holding the following season's strength of schedule constant.[2] These models explain only 5% of the within-team variation in wins over time.

In the context of the model developed by Arel & Tomas (2012), players considering early entry compare team wins in the current season to expect team wins in the

following season when assessing the option value of their next year of eligibility. Given the relatively weak association between team wins this year and wins net year, each additional win this year increases the option value of declaring for the draft this year relative to the option value of next year's eligibility, increasing the likelihood that the player declares early.

In addition to developing this model, Arel & Tomas (2012) analyzed data from the rookie salary scale in place over the 2006–2010 NBA drafts and proxy expected draft position using the outcomes of various mock drafts conducted annually in the run-up to the actual NBA draft. The analysis is noneconometric, showing that early entrants tend to be drafted in the first 10 positions in the NBA draft. This appears consistent with the early exercise of an American put option. Like the papers discussed above, Arel & Tomas (2012) analyzed data from draft outcomes and do not analyze player performance in college.

We investigate decisions by college underclassmen playing NCAA Division I-FBS football in "Power 5" conferences to declare early for the NFL draft or remain in college another year. Our data contain information on college players who declared early for the draft and who decided to remain in college another year. This enables us to assess the validity of the predictions made by the models developed by Groothuis et al. (2007) and Arel & Tomas (2012). Although the models by Groothuis et al. (2007) and Arel & Tomas (2012) focus on the NBA draft, the decision made by football players to declare early for the NFL draft or stay in college another year closely resembles the decision made by college basketball players.

NFL Draft Eligibility Requirements

Our empirical analysis hinges on the institutional characteristics of the NFL draft. This section discusses how the NFL draft changed over time. The NFL first adopted an age eligibility rule of four years beyond high school in 1925, in response to the early exit of Red Grange from the University of Illinois to play for the Chicago Bears during his senior season (Cavezza, 2010). At that time, having satisfied the entry requirement, players were free to sign with any club. The reverse order entry draft was first instituted in 1936 at the urging of Philadelphia Eagles owner, Burt Bell as a way to increase competitive balance.[3] Beginning in 1990, players could apply for "special eligibility" for the draft, meaning they could enter the draft after three seasons beyond the completion of high school instead of four.

Draft eligibility language was first added to the NFL Collective Bargaining Agreement (CBA) in 2006 (Cavezza, 2010). The current rules for eligibility are as follows: To be eligible for the draft, players must have been out of high school for at least three years or must have used up their college eligibility before the start of the next college football season. Underclassmen and players who graduate before using all their college eligibility may request the league's approval to enter the draft early. Players who graduate from high school in December and enroll in

college in January can not count that year toward NFL draft eligibility because the NFL regular season runs until late December or early January.

Note that "redshirt" years in college count toward eligibility for the NFL draft as they represent a year beyond an athlete's high school graduation when an NFL season was played. "Redshirt" years occur when a player practices with the team but does not appear in more than a small number of games in that season; a "redshirt" season does not count toward a player's four years of college eligibility and players can generally only "redshirt" for one season during their college career. "Redshirt" seasons are not restricted to the first year in college.

Based on NFL rules, three types of college football underclassmen are eligible for early entry to the NFL draft: "redshirt" sophomores (players who enrolled in college at an NCAA football program in year t, "redshirted" in year t, and played in years $t + 1$ and $t + 2$); "true" juniors (players who enrolled in college at an NCAA football program in year t and played in years t, $t + 1$, and $t + 2$); and players who played one or more seasons at a community or junior college before transferring to an NCAA college football program.

Since 1990, college players can enter the NFL draft with as little as two seasons of actual college game experience. Despite this, the vast majority of players remain in college beyond the point when they could enter the NFL draft. In this article, we examine the forces that drive the decisions of top-tier college players to either declare early for the draft and forgo their remaining eligibility or remain in the NCAA for one or more additional seasons.

The rule governing NFL draft entry by college football players is unusual even by NCAA standards. College baseball, ice hockey, and beginning in 2018 men's basketball players can retain agents and, in the case of baseball and ice hockey players, even go through the draft without sacrificing their NCAA eligibility, and routinely do so. As of 2018, NCAA basketball players may hire agents approved by the NCAA, though they may not go through the draft without losing their college eligibility.

Since 1994 the NFL draft consists of seven rounds, with 32 teams choosing, in turn, in reverse order (ignoring trades). In total, 224 players get selected over these seven rounds. The total number of players drafted in any draft can be as many as 256 including compensatory draft picks. The number of compensatory picks awarded to teams varies by draft based on a formula comparing players lost in free agency to those signed through free agency by each team that takes into account salary, playing time, and postseason honors for each free agent lost or signed.

In addition, NFL teams routinely offer tryouts to undrafted players and sign these players to contracts. While it is not uncommon for more than a dozen such players to enter training camps, typically only a few end up making a final NFL roster. The NFL caps regular season rosters in 53 and maintains a hard salary cap, limiting the ability of teams to hoard talent from the undrafted market.

Since the current rule formalizing draft eligibility in 1990, the number of players declaring "early" for the NFL draft has steadily increased over time. Figure 1 shows NFL draft outcomes over our sample period. The top line shows the number of

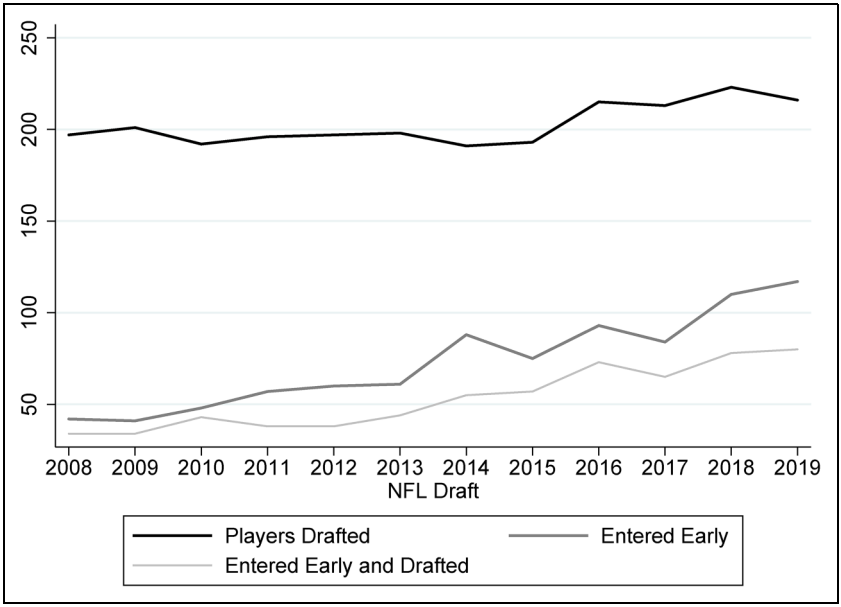

Figure 1. National Football League (NFL) draft outcomes 2008–2019.

drafted players in each draft in our sample. The middle line shows the number of players who declared early. The bottom line shows the number of early entrants who were drafted. In 2019, 117 players in our sample forfeited their remaining NCAA eligibility to enter the NFL draft. Precisely, 68% of these players were drafted, down from an average of > 80% of early entrants drafted in the 2008–2010 drafts. This represents an increase from 110 who declared for the 2018 draft; 71% of the 2018 early entrants were drafted. With 32 teams, the current level of early entrants is enough to fill more than three full rounds of the seven-round NFL draft.

Figure 1 also identifies a limitation of our data. NFL teams drafted between 252 and 256 players over our sample. The exact number varies because of the presence of "compensatory" draft picks awarded to teams who lose players to free agency. We lack complete information on some drafted players because our data source for college football player career information, https://www.sports-reference.com/cfb/, does not include data for all drafted players. These omissions represent FCS players. For example, first round 2019 pick (# 23 overall) Tytus Howard played at FCS Alabama State and second-round 2019 pick Nasir Adderly (# 60 overall) played at the University of Delaware, another FCS school. Neither player appears in the Sports Reference FBS database. No systematic data source for FCS players exists.

The costs of early draft entry to the NCAA are clear as early exit to the professional ranks depletes college athlete labor supply. The most extreme and highly visible example of this is the case of the men's NCAA basketball, wherein "one-and-done,"

players who play only one year of college before turning professional, is the norm for elite college basketball players. Major League Baseball rules allow teams to draft high school baseball players, which also reduces the supply of college athletes.

For players, the risk of declaring early lies in forgoing their final season of pre-professional training and, in so doing, not being selected in the draft or being selected in a later round than if he had stayed in college another season, resulting in a smaller rookie contract. This risk is considerable, since once a player declares for the draft, he cannot return to play in college. The NFL created an advisory committee made up of team and league personnel to give potential early entrants information on whether they would likely be selected and what round, if selected, in 1994.[4] The information provided by this committee to players is not made public, so we cannot account for the impact of this information in the empirical analysis.

Despite this information, > 25% of early entrants go undrafted in the typical NFL draft. These players can sign free agent contracts with any NFL team following the completion of the draft, and can still earn a regular-season roster spot through performance in practice and the preseason. Figure 1 highlights this risk. In the eight most recent drafts, just over 25% of players who declared early were undrafted. An undrafted player who does not make a team roster as a free agent faces few options as a football player. He may also lose out on the possibility of winning an NCAA championship and the opportunity to graduate from college.

The potential benefits to an early entrant include income from a professional contract as opposed to a fourth year in intercollegiate football where compensation is limited to the full cost of college attendance. Particularly for a player coming off a championship season or season in which he excelled individually, declaring early may maximize the value of the first professional contract. Declaring early also eliminates the possibility of serious (perhaps even career ending) injury before reaching the NFL. It also provides additional flexibility to the potential draftee as it relates to both the other players with whom he must compete and the teams that might need his services. For example, consider a college quarterback who knows that the current list of possible draft picks at that position is very limited and also that there are a large number of teams seeking to draft at this position. Being forbidden to enter the draft in that year simply by rule could have large and long-lasting consequences.

The costs to the NFL of early entrants is that such players forfeit one year of pre-professional training. The benefit is that they are assured of receiving at least three years of such training. This arrangement differs from professional baseball. Major League Baseball teams routinely draft players in high school and then pay for at least part of their training in the minor leagues (Winfree & Molitor, 2007). These players have a high failure rate (i.e., never reach the major leagues). In such cases, the major league team invests in training for which they receive no return. In football, the NCAA effectively functions as a free minor league for the NFL. From the standpoint of general human capital training, the NCAA represents a nearly ideal partner for the NFL. While a baseball prospect may be drafted at age 17, a football player would need to be at least 20.

Data and Identification

We collected data on the careers of NCAA Division 1-A (FBS) underclassmen football players eligible for the NFL drafts held after the 2007/08 through 2018/19 college football seasons from https://www.sports-reference.com/cfb/. We end the sample at this season because of the effect of the COVID-19 pandemic on the NFL draft and subsequent college player eligibility changes. This website contains one player entry for each season in which a player appeared in one or more games. Basic information reported for all players include each player's school, his listed position, and the number of games he played in each season. We augmented these data with information about the season performance of each team including conference membership, final season record, and the team strength of schedule from the same website.

The Sports Reference college football player data present several problems when identifying players who were eligible to enter the draft early but did not. First, we only observe the number of seasons in which each player participated in NCAA FBS football and appeared in games. Under current NFL rules, three full college football seasons must elapse between a player's high school graduation and eligibility for the NFL draft. The Sports Reference college football data contains no information on the player's year of high school graduation or age.

The Sports Reference college football data also contains no information about FBS redshirt seasons or seasons played at junior colleges, which also affect NFL draft eligibility. This complicates the identification of players eligible to declare early for the NFL draft in each season. We address this problem by using an exact matching procedure that matches each early draft entrant to one or more players who were eligible but did not enter the draft on using observable characteristics. We describe the matching procedure below.

We augment the Sports Reference college football data set with lists of players who declared early for the NFL draft in each season over the sample period matched by player name, position, and school, downloaded from the NFL website for the 2008–2019 drafts. This provides us with a comprehensive list of all eligible players who actually declared early for each NFL draft. Precisely, 836 (2.4%) players in Sports Reference college football player data appeared on NFL lists of players who declared early for the 2008 through 2019 NFL drafts. We also augment the player data set with the Rivals.com evaluation of each player in his senior year of high school. Rivals.com is a website that evaluates high school athletes who might receive college scholarships. Rivals.com has been ranking high school football players since 2007 and bills itself as "the country's No. 1 authority on college football and basketball recruiting."

Rivals.com rates all high school players believed to have the ability to play FBS college football using a star system where players with zero stars are not rated, players with one star represent the lowest-rated players, and players with five stars represents the highest-rated players. Five star players are among the top 25–30 high school senior football players and have first team All American and first round NFL draft level potential. Four star players are among the top 300 prospects

and have the potential to be drafted high in the NFL draft, three star players are among the top 800 prospects and have the potential to play in the NFL. Importantly, the Rivals star ratings reflect high school statistics and performance as well as other nonmeasurable "soft skill" attributes.

No common player identifier exists to match the Sports Reference college football player data and the Rivals.com data. We match players across these two data sources based on the player's on first name, last name, and the college team they signed with, the only variables common to both data sets. Again, the Sports Reference college football player data does not identify the player's year of high school graduation. After matching the Sports Reference college football player data with the data from Rivals.com using players' names and names of schools they signed with, we are left with an analysis sample containing 5,620 players in total, including 548 players who declared early for the NFL draft (out of the 836 early entrants that appear on the NFL lists) and 5,072 eligible players who did not declare early. The first and second panels on Table 1 contain summary statistics for these two groups. The variable means of early entrants differ from those in the full sample, except for height.

Table 1. Summary Statistics, Underclassmen Eligible to Declare Early.

Variable	Mean	Std. Dev.	Min	Max
Did not declare early—before matching				
Games played season before draft	8.30	4.20	0	15
Team wins season before draft	6.53	3.04	0	15
Team strength of schedule season before draft	0.13	4.29	−10.44	9.78
Rivals stars	2.61	0.76	1	5
Height in inches	73.42	2.51	66	83
Coach change	0.20	0.40	0	1
Observations	5,071			
Declared early				
Games played season before draft	11.77	2.60	1	15
Team wins season before draft	8.84	2.81	1	15
Team strength of schedule season before draft	3.20	3.33	−9.55	9.68
Rivals stars	3.63	0.86	1	5
Height in inches	73.36	2.81	68	80
Coach change	0.14	0.35	0	1
Observations	548			
Did not declare early—after matching				
Games played season before draft	8.43	4.07	1	15
Team wins season before draft	7.11	2.98	0	15
Team strength of schedule season before draft	2.79	3.25	−9.98	9.78
Rivals stars	3.20	0.67	1	5
Height in inches	73.42	2.39	68	79
Coach change	0.20	0.41	0	1
Observations	1,910			

Matched by stars, position, conference, height.

While the NFL lists identify all early draft entrants, these lists cannot help identify players who were eligible to declare but did not declare early for each draft, a natural comparison group for early entrants. Estimation of the relationship between individual and team level observable factors and the decision to declare early for the draft requires identification of a "control" group of players who could have declared early for the NFL draft but did not to compare to the "treated" players who declared early.[5] Using all FBS players, or all players in "Power 5" conferences who did not declare early as the comparison group likely casts too wide a net for an appropriate comparison group. A starting running back at "Power 5" colleges with modest football success like Vanderbilt or Wake Forest may not be a good comparison for a starting running back at "Power 5" football powerhouses like Alabama or Oklahoma. Our exact matching approach addresses this issue.

The matching approach restricts the control group (eligible players who did not declare early) to players with observable characteristics that closely resemble the players who declared early (the treatment group). The observable characteristics include each players' position, conference, height, and ability. As a proxy for ability, we use the player's star rating from Rivals.com reflecting an objective assessment of these players' talent and potential in their senior year of high school.

The exact matching procedure works as follows. We begin with the sample of 5,072 FBS underclassmen eligible to declare early for the NFL draft who did not declare early. We matched the 548 players who declared early to players who were eligible to declare early but did not in our sample based on the number of stars from Rivals.com, the conference, the player competed in, the players' position, and his height in inches. So, for example, we match a Rivals.com three star, six foot tall running back from the Big 10 conference who declared early for the draft to a three star, six foot tall Big 10 conference running back who did not declare early for the draft. The resulting pool of control players contains 1,910 observations and we successfully match 394 observations from the 548 players who declared early, about 70% of the early entrants. This procedure matches some players who declare early with multiple players who did not.

As a robustness check, we also match on the above characteristics in the same season of play. In other words, this process matches a Rivals.com three star, six foot tall running back from the Big 10 conference who declared early for the draft to a three star, six foot tall Big 10 conference running back who did not declare early for the draft in the same season. This more stringent matching approach should eliminate any changes in unobservable incentives to declare early over time. Adding *Season* to the list of our matching variables reduces the number of matched observations in the sample to 191 players who did not enter early and the number of matched early entrants to just 133 players (about 25% of the total number of early entrants).

To increase the number of matched observations in the control group, we loosen the matching process to allow the height of players that we are matching to the early entrants to be within 2 inches of the height of the early entrant. For example, for a player with the height of 73 inches who declared early, we find all the players

playing the same position, in the same conference, in the same season, having the same number of Rivals.com stars with height in the 71–75 inches interval who did not declare early, and add them to our control group. With this alternative approach leaves 750 observations in the control group and matches them to 278 early entrants (about 50% of the group of early entrants) in the treatment group.

The first panel of Table 1 contains summary statistics for the initial sample of 5,072 eligible FBS underclassmen who did not declare early. An average player in this group played 8.30 games in the previous season, his team won average won 6.53 games during the previous season against teams with a strength of schedule equal to 0.13. As for individual statistics, an average player's height is 73.41 inches high and he got an average Rivals.com rating of 2.61 stars. The second panel of Table 1 shows that the eligible players who decided to declare early differ from the sample of eligible players who did not declare early. Early entrants had higher Rivals.com star ratings (3.63), played in more games during the previous season (11.77), and on teams with more wins (8.84) that faced stronger opponents (average strength of schedule 3.63).

The third panel of Table 1 contains summary statistics for the 1,910 players matched to the early entrants on exact characteristics. The variable means for the matched players in the control group are closer to the means for the early entrants in terms of observable characteristics than for the full sample of 5,072 FBS underclassmen who were eligible to declare early but did not. The matched players appeared in more games in the previous season, 8.43 compared to 8.30 games in the full sample. However, both these groups played in fewer games than the early entrants (11.7 games). Matched players played on teams with more wins (7.11) compared to the full sample. This is also closer to the number of wins by teams with players who declared early (8.84). Average Rivals.com stars and team strength of schedule for the matched sample are both much closer to the average for the early entrants.

Note that we only report summary statistics for the matched sample identified using the number of stars, conference, position, and height in Table 1. Summary statistics for the control groups matched on season, and on season and height plus or minus two inches are available by request. We use all three alternative control groups in the regression model results below.

Empirical Analysis

We assess how observable player and college team characteristics reflecting option value and human capital accumulation affect the decision to declare early for the NFL draft. We estimate limited dependent variable models of the form

$$Y_{ipcts} = \gamma_1 PI_p + \gamma_2 SI_s + \gamma_3 CI_c + \gamma_4 TI_t + \beta_1 X_{ipts} + \beta_2 TP_{tcs} + \varepsilon_{ipcts}, \qquad (1)$$

where Y_{ipts} is the binary indicator variable equal to 1 if player i playing position p on college team t in conference c in season s decided to declare early for the NFL draft and equal to zero if the player did not declare early. PI_p is a vector of position

indicator variables (PI_p = defensive back, linebacker, defensive line, offensive line, running back, tight end, wide receiver, quarterback, kicker, and punter) which capture unobservable factors related to each player's position on his decision to declare early for the NFL draft.

SI_s is a vector of season/NFL draft indicator variables (SI_s = 2007, 2008, …, 2018) that capture unobservable factors associated with each FCS season and subsequent NFL draft in the sample. CI_c is a vector of NCAA "Power 5" conference indicator variables (CI_c = ACC, Big 10, Big 12, Southeastern, PAC-12) that capture unobservable conference-level heterogeneity in factors affecting players' decisions to declare early for the NFL draft from that conference. TI_t is a vector of team indicator variables to control for unobservable team-level heterogeneity in factors affecting player's decisions to declare early for the NFL draft associated with each team.

X_{ipts} is a vector of observable player-specific variables. TP_{icts} is a vector of observable, time varying team performance variables for player i's team t in conference c in season s. ε_{ipcts} is an unobservable equation error term capturing all other factors that affect the decision of player i playing position p on team t in conference c in season s decides to declare early for the NFL draft or not in that season. ε_{ipcts} is assumed to be mean zero and possibly heteroscedastic random variable capturing all other factors affecting the decision to declare early for the NFL draft. γ_1, γ_2, γ_3, γ_4, β_1, and β_2 are vectors of unknown parameters to be estimated.

We estimate the unobservable parameters of Equation 1 using the LPM, the ordinary least squares estimator applied to a binary dependent variable. We correct all estimated standard errors for heteroscedasticity using the standard White–Huber "sandwich" correction and cluster-correct all estimated standard errors at the team level. Probit models generate similar results.

We include one player-level variable in X_{ipts}, the number of games played by underclassmen in the season prior to the current NFL draft. We interpret this as a proxy for expected human capital acquisition in the following college season. This reflects the assumption that human capital acquisition takes the form of learning by doing, so players appearing in more games this season acquire more human capital this season, and the assumption of diminishing returns to human capital acquisition from learning by doing. Diminishing returns means that the more games a player appears in this season, the lower will be the expected return from appearing in games next season. In the context of the model developed by Groothuis et al. (2007), appearing in more games this season should increase the likelihood of a player declares early for the upcoming NFL draft, other things equal.

We include two team performance variables in TP_{icts}, including the number of wins by player i's team the season before the NFL draft and the strength of schedule for player i's team during the season before the current NFL draft. We interpret the strength of schedule as a second proxy for the amount of human capital each player acquired in the current season. This variable also reflects learning by doing under the assumption that competing against better opponents leads to more human capital acquisition than competing against lower quality opponents, holding other factors

including games played constant. In the context of the model developed by Groothuis et al. (2007), playing against higher quality opponents this season should increase the likelihood a player declares early for the upcoming NFL draft.

We interpret the team wins variable as a proxy for factors affecting the option value of each players remaining eligibility. Again, a weak association exists between team wins this season and team wins next season, controlling for next season's strength of schedule. This means the average player eligible to declare early after a successful season will not expect that success to continue into the next season. This increases the option value of declaring early relative to staying in college for another year, which, in turn, increases the likelihood a player declares early, based on the model developed by Arel & Tomas (2012). We expect team wins to be correlated with the player performance variable. However, linear regression models can account for correlation between explanatory variables given a large enough sample size.

Also, we include an indicator variable which takes value 1 if there was a change in the head coach of team t during the season t and takes value 0 otherwise. We believe that the change might potentially nudge a player to declare early rather than play under a new coach. We do not include variables capturing any supply-side effects in the NFL draft, for example, the number of other players at each position who might be eligible for the draft, because nether model includes this factor. This means we implicitly assume that roughly the same number of players at each position are eligible for each draft.

Results

We estimate Equation 1 using the three alternative control groups described above using the LPM. Column (1) contains results based on a control group of 1,910 FBS underclassmen who did not declare early for the draft matched to early draft entrants based on number of Rivals.com stars, conference, position, and height. Column (2) contains results based on a control group of 362 FBS underclassmen who did not declare early matched on the above variables in the same season. Column (3) contains results based on a control group of 750 FBS underclassmen matched to early entrants with the relaxed plus/minus 2 inches in height restriction. All these analysis samples contain matched and unmatched early draft entrants and all matched players.

Table 2 contains the results. The results support the predictions in the human capital model of early entry decisions developed by Groothuis et al. (2007). Each additional game played in the season before the draft increases the likelihood that an underclassman declares early by about 3% to 5%, depending on the control group. Greater strength of schedule also affects the early entry decision for the strict control group that matches on season in Column (2). Underclassmen who acquire more human capital in the season before the draft are more likely to declare early.

The results also support the predictions of the option value model developed by Arel & Tomas (2012). Each additional win by an underclassmen's team in the season before the draft increases the likelihood of early entry by 2% to 3%, depending on the control

Table 2. Regression Results—Decision to Declare Early, Matched Control Group (LPM).

	(1)	(2)	(3)
Games played season before draft	0.032***	0.058***	0.053***
	(0.002)	(0.004)	(0.003)
Team wins season before draft	0.024***	0.036***	0.029***
	(0.003)	(0.005)	(0.004)
Team strength of schedule season before draft	0.004	0.003	0.007*
	(0.002)	(0.004)	(0.004)
Coach change	−0.010	0.026	0.011
	(0.019)	(0.042)	(0.032)
Model R^2	0.151	0.286	0.259
Number of observations in treatment group	548	548	548
Number of observations in control group	1,910	362	750
Number of matched observations in treatment	394	133	278
Nonearly entrants matched on	Stars	Stars	Stars
	Position	Position	Position
	Conference	Conference	Conference
	Height	Height	Height ± 2”
		Season	Season

Robust standard errors in parentheses. * 5%, ** 1%, *** \lt 1%.

group. Again, because of the weak association between wins this season and wins next season, each additional win this season decreases the option value of the player's remaining eligibility relative to the current option value, increasing the likelihood of early entry. The coefficient on the Coach Change variable is not statistically significant.

While these results support the predictions of the existing models of early draft entry, this empirical approach has some limitations. In particular, the models contain no individual player-level performance variables, because of a lack of consistent performance variables across different football position.[6] To assess the importance of excluding individual player performance measures from the empirical model, we undertake a robustness check focusing on a single position—quarterback. Focusing on a single position, and in this case, the most important player on a football team allows for the inclusion of variables reflecting individual player performance.

Results for Quarterbacks

Human capital acquisition and the option value of a player's remaining college eligibility clearly affect the decision to enter the NFL draft early. This supports both the human capital model perspective and the real option perspective. The results in the previous section used games played and team strength of schedule as proxy variables for human capital acquisition, because we lack other consistent across-player performance variables for the entire sample. To assess the importance of specific player performance variables related to human capital acquisition on the decision

to enter the NFL draft early, we estimate Equation 1 using only data for underclassmen college quarterbacks. This allows for the inclusion of specific performance variables related to human capital in the empirical model.

We use three specific performance measures for quarterbacks in the season before the current NFL draft: passing yards, touchdown passes thrown, and pass attempts. These three variables reflect human capital acquisition in terms of learning by doing; more pass attempts, yards, and touchdowns in a season means more experience for a quarterback in that season. "Dual Threat" NFL quarterbacks integrated into both the running and passing game became more common over the sample period. This could make quarterback rushing ability an important component of the decision to declare early for the draft. Unfortunately, differences in the offensive schemes used by college football teams make it difficult to distinguish between "Dual Threat" college quarterbacks and other college quarterbacks using rushing yards.

Unlike in the NFL, a number of college teams employ a run-option offensive scheme. In this scheme, the quarterback represents an additional option for running the football. Option quarterbacks typically attempt relatively low numbers of passes in games, making them unattractive draftees for NFL teams. They also amass rushing yards similar to "Dual Threat" quarterbacks. For example, the two highest quarterback rushing seasons in the sample were by Lamar Jackson and Ahmad Bradshaw. While Jackson is clearly a "Dual Threat" NFL quarterback, Bradshaw was an option offense quarterback who did not play in the NFL and played four years at the U.S. Military Academy. He would not have considered turning pro early.

These variables provide more variability in quarterback human capital acquisition than the games played variable used above, providing an important robustness check. These variables also reflect teammate performance, since quarterbacks on teams with better offensive linemen and receivers will pass for more yards and more touchdowns. Including team wins will partially control for these effects. Like the regression models above, we restrict the sample to quarterbacks who declared early for the NFL draft, or could have declared early based on their years in the sample.

The quarterback subsample contains data on the final college season for 371 quarterbacks in their second or third year of FBS play over the period 2007/08-2018/19. Table 3 contains sample means by draft entry status for quarterback and team performance variables. The quarterbacks who declared early for the draft are elite players who perform substantially better than quarterbacks who do not declare early for the draft. Rivals rated the average early entrant at 3.81 stars and all other quarterbacks at 2.80 stars, a roughly one-star difference. Early entrants are nearly an inch taller than the other quarterbacks in the sample. Height represents the desirable quarterback characteristics, since taller players can see better over offensive and defensive linemen.

Both Rivals.com high school stars and height represent controls for quarterback ability unrelated to human capital acquisition during college. Including these variables in regression models explaining the decision to declare early for the NFL draft should sharpen estimates of factors reflecting the human capital acquisition.

Table 3. Sample Means—Quarterback Performance by Draft Status.

	All other QBs	Early entrants
Games played season before draft	6.55	11.63
Team wins season before draft	6.17	8.66
Team strength of schedule season before draft	−0.03	2.79
Rivals stars	2.80	3.81
Height in inches	74.7	75.6
Passing yards season before draft	741	3,054
Touchdown passes season before draft	5.07	23.75
Pass attempts season before draft	107	374

The average quarterback declaring early played in more games, attempted and completed more passes, threw more touchdown passes, and more interceptions than other quarterbacks in the sample. In terms of team performance, the average quarterback declaring early for the draft played on a more successful team, and on a team that faced higher quality opponents, in the year prior to the NFL draft.

We estimate Equation 1 adding the three quarterback performance variables as explanatory variables separately. Following the full sample results above, we also include the Rivals.com high school star ranking, height, games played, team wins, and team strength of schedule in season before each NFL draft for each quarterback in the sample as explanatory variables in all models. Standard errors are corrected for heteroscedasticity using the standard White-Huber "sandwich" correction and cluster corrected at the team level. Results appear in Table 4. Column (1) contains results including passing yards, column (2) touchdown passes, and column (3) pass attempts. These variables should reveal more about human capital acquisition over the course of the season than the number of games played.

Results in Table 4 show that more Rivals.com star ratings substantially increase the likelihood of a quarterback declaring early for the draft. Each additional Rivals.com star increases the likelihood of declaring early by about 4 percentage points across all three models. High school star rankings contain important information about the likelihood a quarterback will declare early for the NFL draft, holding constant college human capital acquisition. Taller players are more likely to declare early, confirming the idea that height represents a positive quality for quarterbacks.

Games played in the previous season have no statistical relationship to the decision to declare early, unlike the full sample results in the previous section, when controlling for other variables reflecting quarterback performance. While passing yards, touchdown passes, and pass attempts provide more explanatory power than games played, these results suggest that the games played variable in the full sample represents an adequate human capital acquisition proxy.

Team wins in the previous season has a positive and statistically significant parameter estimate in two of the three models in Table 4. Again, we interpret this as evidence supporting the put option model developed by Arel & Tomas (2012).

Table 4. Regression Results—Decision by Quarterbacks to Declare Early (LPM).

	(1)	(2)	(3)
Rivals stars	0.039*	0.039*	0.040*
	(2.23)	(2.21)	(2.23)
Height in inches	0.023**	0.025***	0.023**
	(3.13)	(3.43)	(3.12)
Games played season before draft	−0.0071	−0.0055	−0.0049
	(−1.80)	(−1.35)	(−1.28)
Team wins season before draft	0.0091*	0.0064	0.012**
	(2.29)	(1.61)	(2.98)
Team strength of schedule season before draft	0.010	0.0098	0.010
	(1.38)	(1.31)	(1.35)
Passing yards season before draft (00s)	0.013***		
	(6.39)		
Touchdown passes season before draft		0.017***	
		(5.84)	
Pass attempts season before draft (00s)			0.095***
			(6.12)
Observations	371	371	371
R^2	0.403	0.412	0.376

Robust standard errors in parentheses. * 5%, ** 1%, *** \lt 1%.

Like the full sample results, the parameter estimate on the team strength of schedule in the previous season variable cannot be statistically distinguished from zero.

All three of the quarterback performance variables in the previous season have positive, statistically significant parameters. These results generally support the importance of human capital acquisition in the decision to declare early for the draft in the sense that quarterbacks with more human capital perform better than the quarterbacks with less human capital. Note that we express all three of these passing variables in hundreds. The parameter estimate on the passing yards variable implies that each additional 100 yards passing the previous season increases the probability that a quarterback with eligibility remaining declares early for the draft by 1.3 percentage points.

The coefficient on the pass attempts variable exceeds the parameter estimates on the other two variables by a large margin. Each additional 100 passing attempts in the previous season increases the likelihood that an underclassmen quarterback declared early for the draft by 9.5 percentage points. This makes intuitive sense. Passing represents an important job component for quarterbacks, and FBS quarterbacks attempting more passes acquire more human capital related to this activity; in turn, under the assumption of diminishing returns to learning by doing, they would expect to gain less such human capital in the following season, increasing the likelihood of declaring for the draft this year, other things equal.

Taken together, the results in Table 4 support the results from the full sample, and also provide new information about the decision to declare early for the draft. The

quarterback's early draft entry decisions reflect the same factors as in the full sample, human capital acquisition and increased exposure from playing on a winning team that likely increases player's draft position, a proxy for option value. More detailed data on the specific type of quarterback learning by doing contains more explanatory power than games played, confirming the importance of human capital acquisition.

Conclusions

Many talented college athletes face a high-stakes "stay or go pro" decision in terms of declaring for entry drafts conducted by North American professional sports leagues. College football and basketball players entering the NFL and NBA draft early forfeit all remaining college eligibility. Entry draft position plays a critical role in determining early-career earnings, so foregoing additional college training can possibly reduce career professional earnings, or in the case of undrafted early entrants, eliminate the possibility of a professional career. While the substantial prior research focused on analyzing draft outcomes and postdraft career length and performance, no previous empirical research addressed the decision to declare early from the perspective of college players faced with this high-stakes decision.

We analyzed the decision to declare early for the NFL draft faced by FBS underclassmen with remaining eligibility and identified observable characteristics associated with the decision to declare early. The results broadly support both existing theoretical models addressing the decision to declare early for entry drafts. Human capital, emphasized in the model developed by Groothuis et al. (2007), and the option value associated with turning pro early, emphasized in the model developed by Arel & Tomas (2012), both models explain the early entry in the full sample and quarterback subsample.

The results also highlight the importance of both training and exposure during college relative to ability or human capital prior to players entering college. The Rivals.com stars reflect high school performance. From Table 4, each additional Rivals.com star increases the likelihood a quarterback declares early for the draft by about 4 percentage points. The parameter estimates on the performance measures indicate that passing for another 300 yards over the course of the previous season, or throwing an additional three touchdown passes over the course of the previous season, lead to similar increases in the likelihood a college quarterback with remaining eligibility declares early for the NFL draft.

Testing the predictions of economic models represents an important contribution to empirical economic research. We undertake the first empirical tests for the economic models developed by Groothuis et al. (2007) and Arel & Tomas (2012) and find support for both. Empirical support can help to refine economic models. As the two existing models of the early draft entrance are not nested, our support for both implies that these models need to be extended in general directions to make

them comparable. The human capital approach could be extended to include a pure option value component, or vice versa.

Several important changes occurred in the American football environment after the end of our analysis sample. First, two new minor professional football leagues, XFL and the United States Football League, started play in 2020 and 2022, respectively. While not paying salaries as large as NFL teams, teams in these leagues represent potential employers for early draft entrants that do not make an NFL team roster, increasing the benefits of declaring early. Second, the NCAA changed its rules to allow college athletes to earn money from name, image, and likeness (NIL) rights while still maintaining college eligibility starting in 2021. The ability to earn NIL money increased the value of remaining in college another year. Third, the NFL granted all players an extra year of eligibility during the COVID-19 pandemic. These events occurred after our sample period, but future research should address the impact of these changes on the decision to declare early.

Finally, the opening of two new minor professional football leagues in 2020 and 2021, and the changes in NCAA regulations about NIL earnings, likely had important impacts on the decision by college football players to declare early for the NFL draft. These changes occurred after our analysis sample period. They also represent ideal difference-in-differences experiments for future research on this topic using updated data from the postevent period.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. Winfree & Molitor (2007) analyzed decisions made by high school baseball players about going to college to play baseball or playing minor league baseball. This does not apply to football players, as they cannot play professionally immediately after graduating from high school.

2. The parameter estimate on the current season strength of schedule has the expected negative and statistically significant sign. These results are available by request.
3. Pro Football Hall of Fame (n.d.). The four-year rule governed all NFL drafts up to 1989.
4. See K. Seifert, How the NFL advises college underclassmen on entering the draft. ESPN.com, NFL Nation. January 17, 2017. At http://www.espn.com/blog/nflnation/post/_/id/229070/how-the-nfl-advises-college-underclassmen-on-entering-the-draft. Accessed May 30, 2019.
5. Note that we adopt the standard causal inference terminology of "treated" and "control" players for clarity. We recognize that declaring early for the draft is not randomly assigned and that we do not have a natural experiment.
6. While some position-independent performance measures exist for football players, for example, wins above average, these data are not used in the research literature. This likely reflects the difficulty of defining the performance of an "average" college football player.

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